

182. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

182.1 Overview

This clause specifies the 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 Physical Medium Dependent (PMD) types together with the single-mode fiber medium. The optical signal generated by this PMD type is modulated using a 4-level pulse amplitude modulation (PAM4) format. These PMD types provide point-to-point 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet links over one, two, four, or eight pairs, respectively, of single-mode fiber, with a reach of up to at least 2 km.

When forming a complete Physical Layer, a PMD shall be connected to the appropriate PMA as shown in Table 182–1 through Table 182–4, to the medium through the medium dependent interface (MDI).

Table 182–1—Physical Layer clauses associated with the 200GBASE-DR1-2 PMD

Associated clause	200GBASE-DR1-2
90—Time Synchronization	Optional
117—200 Gb/s RS	Required
117—200GMII ^a	Optional
118—200GMII Extender	Optional
119—200GBASE-R PCS	Required
120—200GBASE-R BM-PMA	Conditional ^b
120B—200GAUI-8 C2C ^c	Optional ^d
120C—200GAUI-8 C2M ^c	Optional ^d
120D—200GAUI-4 C2C ^c	Optional ^d
120E—200GAUI-4 C2M ^c	Optional ^d
120F—200GAUI-2 C2C	Optional ^d
120G—200GAUI-2 C2M	Optional ^d
176—200GBASE-R SM-PMA	Required ^b
176C—200GAUI-1 C2C	Optional ^d
176D—200GAUI-1 C2M	Optional ^d
177—200GBASE-R Inner FEC	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

^b If a 200GAUI-n is implemented in a PHY, additional 200GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.5.1.

^c When implemented in a 200GBASE-DR1-2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 200GAUI-n may be instantiated within a 200GBASE-DR1-2 PHY as described in 176B.5.1.

Table 182–2—Physical Layer clauses associated with the 400GBASE-DR2-2 PMD

Associated clause	400GBASE-DR2-2
90—Time Synchronization	Optional
117—400 Gb/s RS	Required
117—400GMII ^a	Optional
118—400GMII Extender	Optional
119—400GBASE-R PCS	Required
120—400GBASE-R BM-PMA	Conditional ^b
120B—400GAUI-16 C2C ^c	Optional ^d
120C—400GAUI-16 C2M ^c	Optional ^d
120D—400GAUI-8 C2C ^c	Optional ^d
120E—400GAUI-8 C2M ^c	Optional ^d
120F—400GAUI-4 C2C	Optional ^d
120G—400GAUI-4 C2M	Optional ^d
176—400GBASE-R SM-PMA	Required ^b
176C—400GAUI-2 C2C	Optional ^d
176D—400GAUI-2 C2M	Optional ^d
177—400GBASE-R Inner FEC	Required

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

^b If a 400GAUI-n is implemented in a PHY, additional 400GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.6.1.

^c When implemented in a 400GBASE-DR2-2 PHY, the signaling rate range is constrained as specified in 120.1.4.

^d One or two 400GAUI-n may be instantiated within a 400GBASE-DR2-2 PHY as described in 176B.6.1.

Table 182–3—Physical Layer clauses associated with the 800GBASE-DR4-2 PMD

Associated clause	800GBASE-DR4-2
90—Time Synchronization	Optional
120F—800GAUI-8 C2C	Optional ^a
120G—800GAUI-8 C2M	Optional ^a
170—800 Gb/s RS	Required
170—800GMII ^b	Optional
171—800GMII Extender	Optional
172—800GBASE-R PCS	Required
173—800GBASE-R BM-PMA	Conditional ^c
176—800GBASE-R SM-PMA	Required ^c
176C—800GAUI-4 C2C	Optional ^a
176D—800GAUI-4 C2M	Optional ^a
177—800GBASE-R Inner FEC	Required

^a One or two 800GAUI-n may be instantiated within a 800GBASE-DR4-2 PHY as described in 176B.7.1.

^b The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

^c If a 800GAUI-n is implemented in a PHY, additional 800GBASE-R BM-PMA or SM-PMA sublayers are required according to 176B.7.1.

Table 182–4—Physical Layer clauses associated with the 1.6TBASE-DR8-2 PMD

Associated clause	1.6TBASE-DR8-2
90—Time Synchronization	Optional
120F—1.6TAUI-16 C2C	Optional ^a
120G—1.6TAUI-16 C2M	Optional ^a
170—1.6Tb/s RS	Required
170—1.6TMII ^b	Optional
171—1.6TMII Extender	Optional
175—1.6TBASE-R PCS	Required
176—1.6TBASE-R SM-PMA	Required ^c
176C—1.6TAUI-8 C2C	Optional ^a
176D—1.6TAUI-8 C2M	Optional ^a
177—1.6TBASE-R Inner FEC	Required

^a One or two 1.6TAUI-n may be instantiated within a 1.6TBASE-DR8-2 PHY as described in 176B.8.1.

^b The 1.6TMII is an optional interface. However, if the 1.6TMII is not implemented, a conforming implementation behaves functionally as though the RS and 1.6TMII were present.

^c If a 1.6TAUI-n is implemented in a PHY, additional 1.6TBASE-R SM-PMA sublayers are required according to 176B.8.1.

Figure 182–1 shows the relationship of the PMD and MDI (shown shaded) along with other Ethernet sublayers to the ISO/IEC Open System Interconnection (OSI) reference model. 200 Gb/s Ethernet and 400 Gb/s Ethernet are introduced in Clause 116 and the purpose of each PHY sublayer is summarized in 116.2. 800 Gb/s Ethernet is introduced in Clause 169 and the purpose of each PHY sublayer is summarized in 169.2. 1.6 Tb/s Ethernet is introduced in Clause 174 and the purpose of each PHY sublayer is summarized in 174.2.

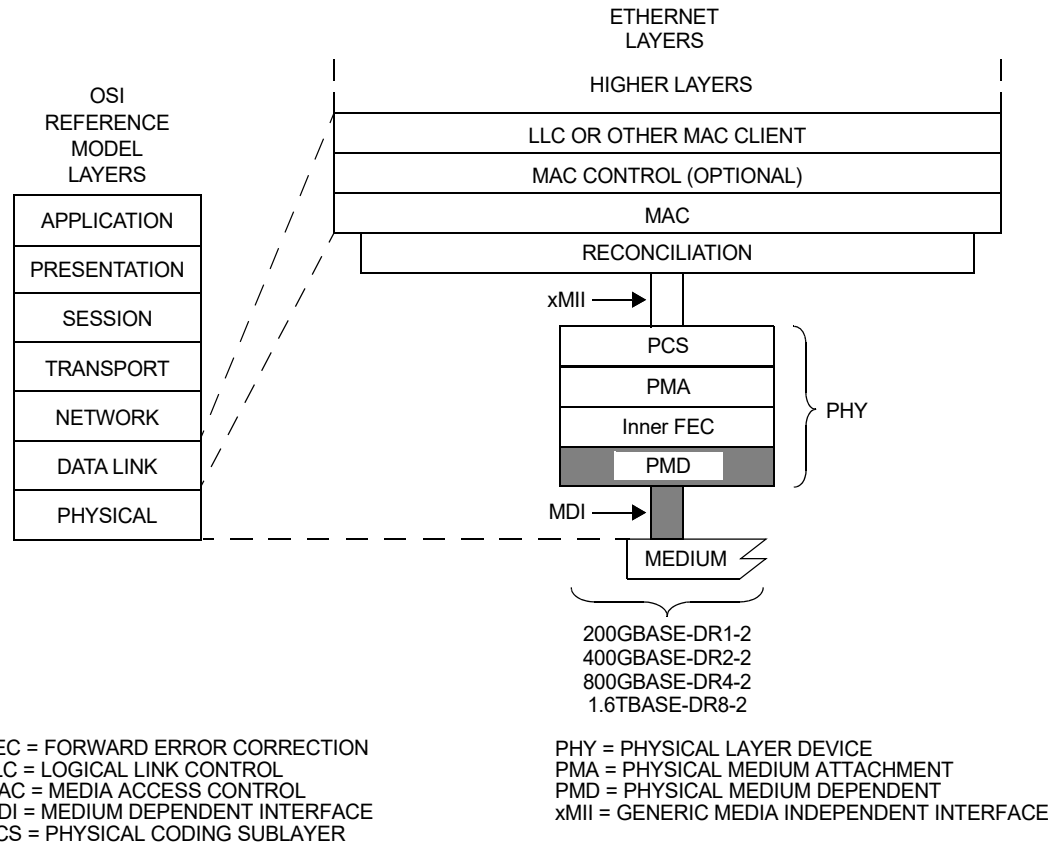


Figure 182-1—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMD relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

182.2 Error ratio allocation

A complete PHY is expected to meet the frame loss ratio specifications in 174A.6.

With a compliant input signal, a PMD receiver is expected to meet the block error ratio (BLER) of 1.45×10^{-11} (see 174A.6), measured at the PMA nearest to the PMD using the method described in 174A.9, with BER_{added} equal to 6.4×10^{-5} .

With a compliant input signal, a PHY receiver is expected to meet the BLER of 1.45×10^{-11} (see 174A.6), measured at the PCS using the method described in 174A.11, with BER_{added} equal to 3.2×10^{-5} .

182.3 PMD service interface

This subclause specifies the services provided by the PMD. The service interface for this PMD is described in an abstract manner and does not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMD and the PMD client (Inner FEC immediately above the PMD). The PMD translates the encoded data to and from signals suitable for the specified medium.

The PMD service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and 174.3 for 1.6TBASE-R, except that the rx_symbol parameter is redefined.

For PMD:IS_UNITDATA_i.request primitive in the transmit direction and PMD:IS_UNITDATA_i.indication primitive in the receive direction, where $i = 0$ to $n-1$, the tx_symbol and rx_symbol parameters are parallel PAM4 symbol streams with a nominal signaling rate of 113.4375 GBd. The number of parallel streams, n , is 1 for 200GBASE-DR1-2, 2 for 400GBASE-DR2-2, 4 for 800GBASE-DR4-2, and 8 for 1.6TBASE-DR8-2. Each of the tx_symbol parameters takes on one of four (PAM4) values: 0, 1, 2, or 3. Each of the rx_symbol parameters takes a set of four or more implementation-specific values to support the Inner FEC soft-decision decoder (see 177.5.4).

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive (see 116.3.3.3) corresponds to the variable rts_status of the ready-to-send (RTS) function (see 182.5.12), as defined in 178B.8.2.1. When SIGNAL_OK is either IN_PROGRESS or FAIL, the rx_symbol parameters of PMD:IS_UNITDATA_i.indication primitive on all lanes are unspecified.

The PMD receives signal status information from the PMD client using the PMD:IS_SIGNAL.request primitive (see 116.3.3.4).

182.4 Delay and Skew

182.4.1 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the 200GBASE-DR1-2 PMD including 2 m of fiber in one direction shall be no more than 14 848 bit times (29 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 400GBASE-DR2-2 PMD including 2 m of fiber in one direction shall be no more than 29 696 bit times (58 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 800GBASE-DR4-2 PMD including 2 m of fiber in one direction shall be no more than 59 392 bit times (116 pause_quanta or 74.24 ns).

The sum of the transmit and receive delays at one end of the link contributed by the 1.6TBASE-DR8-2 PMD including 2 m of fiber in one direction shall be no more than 118 784 bit times (232 pause_quanta or 74.24 ns).

Descriptions of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 for 200GBASE-DR1-2 and 400GBASE-DR2-2, in 169.4 for 800GBASE-DR4-2, and in 174.4 for 1.6TBASE-DR8-2.

182.4.2 Skew constraints

The Skew (relative delay between the lanes) and Skew Variation are kept within limits so that the information on the lanes can be reassembled by the PCS.

182.4.2.1 Skew constraints for 200GBASE-DR1-2 and 400GBASE-DR2-2

Skew and Skew Variation for 200GBASE-DR1-2 and 400GBASE-DR2-2 are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–4 and Figure 116–5. For 200GBASE-DR1-2, the signal at these points represents a single symbol stream, so there is no Skew Variation.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 200GBASE-DR1-2 and 400GBASE-DR2-2 see 116.5.

182.4.2.2 Skew constraints for 800GBASE-DR4-2 and 1.6TBASE-DR8-2

Skew and Skew Variation for 800GBASE-DR4-2 are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–4 and Figure 169–5.

Skew and Skew Variation for 1.6TBASE-DR8-2 are defined in 174.5 and specified at the points SP1 to SP6 shown in Figure 174–5 and Figure 174–6.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation for 800GBASE-DR4-2 see 169.5 and for 1.6TBASE-DR8-2 see 174.5.

182.5 PMD functional specifications

The 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMDs perform the Transmit and Receive functions, which convey data between the PMD service interface and the MDI.

182.5.1 Block diagram

The block diagram for 800GBASE-DR4-2 is shown in Figure 182–2.

The block diagram for 200GBASE-DR1-2 is equivalent to Figure 182–2, but for one lane per direction, instead of four.

The block diagram for 400GBASE-DR2-2 is equivalent to Figure 182–2, but for two lanes per direction, instead of four.

The block diagram for 1.6TBASE-DR8-2 is equivalent to Figure 182–2, but for eight lanes per direction, instead of four.

The inter-sublayer link training (ILT) and RTS functions indicated in Figure 182–2 are defined in Annex 178B.

For purposes of system conformance, the PMD sublayer is standardized at the points described in this subclause. The optical transmit signal is defined at the output end of a single-mode fiber patch cord (TP2), between 2 m and 5 m in length. Unless specified otherwise, all transmitter measurements and tests defined in 182.9 are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI (see 182.8.3). Unless specified otherwise, all receiver measurements and tests defined in 182.9 are made at TP3.

While TP2 and the patch cord used for transmitter measurements and tests are shown in Figure 182–2, they are not part of the medium. The transmit MDI is connected to the medium for use, or to the patch cord and TP2 for transmitter measurements and tests.

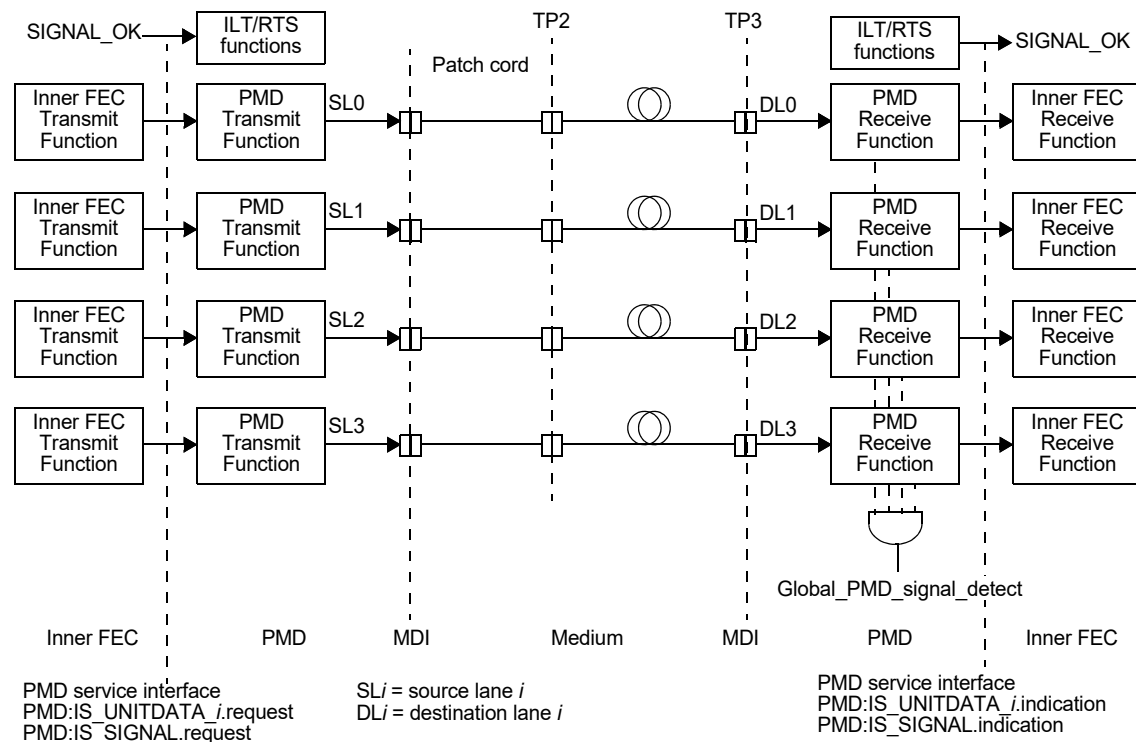


Figure 182–2—Block diagram for 800GBASE-DR4-2 transmit/receive paths

182.5.2 PMD transmit function

The PMD transmit function has two operating modes: DATA and TRAINING. The operating mode is controlled by the tx_mode variable of the ILT function (see 182.5.12): it is DATA when tx_mode = DATA, and TRAINING otherwise.

When operating in DATA mode, the PMD transmit function shall convert the n symbol streams requested by the PMD service interface PMD:IS_UNITDATA_0.request primitive to PMD:IS_UNITDATA_ n -1.request primitive into n separate optical signals. The n optical signals shall then be delivered to the MDI, which contains n parallel light paths for transmit, according to the transmit optical specifications in this clause.

The polarity of the PMD output on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI output, in order from lowest to highest, correspond to tx_symbol values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI output, in order from highest to lowest, correspond to tx_symbol values zero, one, two, and three, respectively.

If training is enabled, either normal or inverted output polarity may be used, since the PMD receiver in the peer interface detects the polarity and corrects it if necessary (see 178B.7.7). If training is disabled, the output polarity of each lane shall be normal. Output polarity may be controlled by management in an implementation dependent manner.

When operating in TRAINING mode, the PAM4 symbol stream on each lane is provided by the ILT function (see 178B.7).

182.5.3 PMD receive function

The PMD Receive function shall convert the n parallel optical signals received from the MDI into separate symbol streams for delivery to the PMD service interface using the PMD:IS_UNITDATA_0.indication primitive to PMD:IS_UNITDATA_ n -1.indication primitive, all according to the receive optical specifications in this clause.

The polarity of the PMD input on each of the lanes is either normal or inverted. When normal, the four optical power levels at the MDI input, in order from lowest to highest, correspond to tx_symbol values zero, one, two, and three, respectively. When inverted, the four optical power levels at the MDI input, in order from highest to lowest, correspond to tx_symbol values zero, one, two, and three, respectively.

If training is enabled, the PMD shall detect the polarity during training and correct it if necessary (see 178B.7.7). If training is disabled, the input polarity of each lane shall be normal. Input polarity may be controlled by management in an implementation dependent manner.

182.5.4 PMD global signal detect function

The variable Global_PMD_signal_detect is a global indicator of the presence of optical signals on all n lanes. For the 200GBASE-DR1-2 PMD, the variable Global_PMD_signal_detect is set to PMD_signal_detect_0 (see 182.5.5). For the 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMDs, the variable Global_PMD_signal_detect is set to the logical-and of PMD_signal_detect_ i from all of the PMD lanes.

182.5.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate sufficient optical power is detected at the receiver input on each lane. For each lane i , where i is in the range 0 to $n-1$, a receiver shall set the variable PMD_signal_detect_ i to either 0 or 1 according to the criteria in Table 182-5.

Various implementations of the signal detect function are permitted by this standard. This standard imposes no response time requirements on the generation of the `PMD_signal_detect_i` variable.

Implementations need to provide adequate margin between the input optical power level at which the `PMD_signal_detect_i` variable is set to 1 and the inherent noise level of the PMD, including the effects of crosstalk, power supply noise, etc.

Table 182–5—Signal detection criteria for each lane

Receive conditions	<code>PMD_signal_detect_i</code>
Average optical power at TP3 ≤ -15 dBm	0
[The input signal is a compliant 200GBASE-R, 400GBASE-R, 800GBASE-R or 1.6TBASE-R signal] ^a AND [Optical power at TP3 $\geq$ average receive power, each lane (min) in Table 182–8]	1
All other conditions	Unspecified, may be 0 or 1

^a This is a condition for testing of the receiver, not a requirement for the receiver to detect that the signal is valid.

182.5.6 PMD reset function

If the variable `PMD_reset` is asserted, the PMD shall be reset as defined in 45.2.1.1.1.

PMD reset shall reset the ILT and RTS functions associated with the PMD (see 182.5.12).

182.5.7 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional and allows all of the optical transmitters to be disabled.

When the `Global_PMD_transmit_disable` variable is set to one, this function shall turn off all of the optical transmitters so that each transmitter meets the requirements of the average launch power of the OFF transmitter in Table 182–7.

If a `PMD_fault` is detected, then the PMD may set the `Global_PMD_transmit_disable` variable to one, turning off the optical transmitter in each lane.

182.5.8 PMD lane-by-lane transmit disable function

The PMD lane-by-lane transmit disable function allows the optical transmitters in each lane to be selectively disabled. This function is used by the ILT function. It may optionally be provided through control variables `PMD_transmit_disable_i`.

If either `PMD_transmit_disable_i` or the variable `tx_disable` in lane i of the ILT function (see 178B.8.2.1) is set to one (where i represents the lane number in the range $0:n-1$), this function shall turn off the associated optical transmitter so that the transmitter meets the requirements of the average launch power of the OFF transmitter in Table 182–7.

If a `PMD_fault` is detected, then the PMD may set each `PMD_transmit_disable_i` variable to one, turning off the optical transmitter in each lane.

182.5.9 PMD fault function (optional)

If the PMD has detected a local fault on any of the transmit or receive paths, the PMD shall set PMD_fault to one.

182.5.10 PMD transmit fault function (optional)

If the PMD has detected a local fault on any transmit lane, the PMD shall set the PMD_transmit_fault variable to one.

182.5.11 PMD receive fault function (optional)

If the PMD has detected a local fault on any receive lane, the PMD shall set the PMD_receive_fault variable to one.

182.5.12 Autonomous path startup (APSU) functions

A PMD shall provide the RTS function and the ILT function with a training frame of O1 format, specified in Annex 178B.

When the variable mr_training_enable is true, the ILT function is used to request changes to the peer transmitter state (modulation, training pattern, and precoder state), indicate the receiver state, and coordinate the transition to DATA mode.

When tx_mode = LOCAL_PATTERN (see 178B.8.3.1), the PMD transmits PRBS31 encoded by Inner FEC (see 177.6.1.1).

The default value of max_wait_timer_duration is 60, corresponding to a duration of 60 seconds for max_wait_timer (see 178B.8.3.3).

182.6 Lane assignments

In order to support the ILT function as described in 182.5.12, optical lane assignments (within a group of transmit or receive lanes) for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 are specified. The positioning and ordering of transmit and receive lanes at the MDI is specified in 182.9.

Each fiber between the transmitter of one PHY and the receiver of another PHY shall connect to the same signal lane number. For example, a fiber connects SL1 at the transmitting end to DL1 at the receiving end.

182.7 PMD to MDI optical specifications

The operating range for the 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 PMDs is defined in Table 182–6. A 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 compliant PMD operates on type G.652.D, G.657.A1, or G.657.A2 single-mode fibers according to the specifications defined in Table 182–11. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 800GBASE-DR4-2 PMD operating at 2.5 km meets the operating range requirement of 2 m to 2 km).

Table 182–6—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 operating range

PMD type	Required operating range
200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2	2 m to 2 km

182.7.1 Transmitter optical specifications

A transmitter shall meet the specifications defined in Table 182–7 per the definitions in 182.9.

Table 182–7—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 transmit characteristics

Description	Defined in	200GBASE-DR1-2	400GBASE-DR2-2 800GBASE-DR4-2 1.6TBASE-DR8-2	Unit
Signaling rate, each lane (range)	—	113.4375 ± 50 ppm		GBd
Modulation format	—	PAM4		—
Lane wavelength (range)	182.9.3	1304.5 to 1317.5		nm
Side-mode suppression ratio (SMSR), each lane (min)	182.9.3	30		dB
Average launch power, each lane (max)	182.9.4	4		dBm
Average launch power, each lane ^a (min)	182.9.4	−3.1 ^b		dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (max)	182.9.5	4.2		dBm
Outer optical modulation amplitude (OMA _{outer}), each lane (min) for max(TECQ, TDECQ) < 0.9 dB for 0.9 dB ≤ max(TECQ, TDECQ) ≤ 3.4 dB	182.9.5	−0.1 −1 + max(TECQ, TDECQ)		dBm dBm
Transmitter and dispersion eye closure for PAM4 (TDECQ), each lane (max)	182.9.6	3.4		dB
Transmitter eye closure for PAM4 (TECQ), each lane (max)	182.9.7	3.4		dB
TDECQ − TECQ , each lane (max)	—	2.5		dB
Transmitter functional symbol error histogram, each lane (max)	182.9.8	See Table 180–17		—
Transmitter overshoot and undershoot, each lane (max)	182.9.9	27		%

Table 182–7—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 transmit characteristics (*continued*)

Description	Defined in	200GBASE-DR1-2	400GBASE-DR2-2 800GBASE-DR4-2 1.6TBASE-DR8-2	Unit
Transmitter power excursion, each lane (max)	182.9.10	2.9		dBm
Extinction ratio, each lane (min)	182.9.11	3.5		dB
Transmitter transition time, each lane (max)	182.9.12	8		ps
Average launch power of OFF transmitter, each lane (max)	—	–15		dBm
RIN _x OMA, each lane (max) x = 17.1 x = 21.4	182.9.13	–139 —	— –139	dB/Hz dB/Hz
Optical return loss tolerance, each lane (max)	182.9.13	17.1	21.4	dB
Transmitter reflectance ^c , each lane (max)	—	–26		dB

^a Average launch power, each lane (min) is not the principal indicator of signal strength. A transmitter with launch power below this value cannot be compliant; however, a value above this does not ensure compliance.

^b An average launch power of –3.1 dBm corresponds to an OMA of –0.1 dBm with an infinite extinction ratio.

^c Transmitter reflectance is defined looking into the transmitter.

The values for OMA_{outer} each lane (min) in Table 182–7 vary with max (TECQ, TDECQ). The relationships are illustrated in Figure 182–3 along with the values for OMA_{outer} each lane (max).

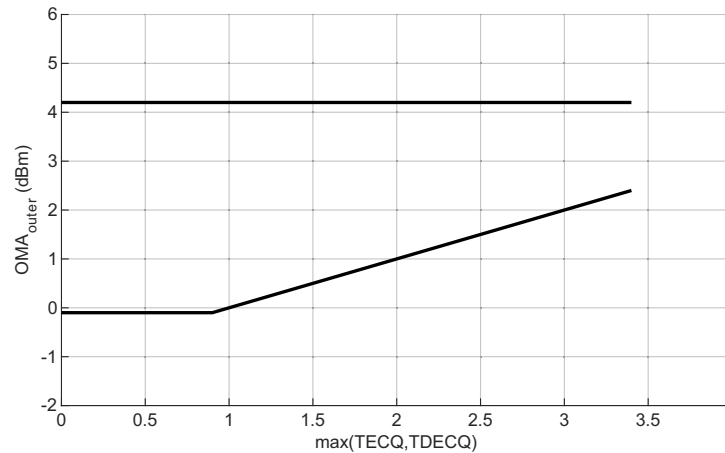


Figure 182–3—OMA_{outer} each lane (max) and OMA_{outer} each lane (min) versus max (TECQ, TDECQ)

182.7.2 Receive optical specifications

A receiver shall meet the specifications defined in Table 182–8 per the definitions in 182.9.

Table 182–8—200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 receive characteristics

Description	Defined in	Value	Unit
Signaling rate, each lane (range)	—	113.4375 ± 50 ppm	GBd
Modulation format	—	PAM4	—
Lane wavelengths (range)	182.9.3	1304.5 to 1317.5	nm
Damage threshold ^a , each lane	—	5	dBm
Average receive power, each lane (max)	—	4	dBm
Average receive power, each lane ^b (min)	—	−7.1	dBm
Receive power (OMA _{outer}), each lane (max)	—	4.2	dBm
Receiver reflectance (max)	—	−26	dB
Receiver sensitivity (OMA _{outer}), each lane (max) for TECQ < 0.9 dB for 0.9 dB ≤ TECQ ≤ SECQ	182.9.14	−4.4 −5.3 + TECQ	dBm
Stressed receiver sensitivity (OMA _{outer}), each lane ^c (max)	182.9.15	−1.9	dBm
Conditions of stressed receiver sensitivity test ^d :			
Stressed eye closure for PAM4 (SECQ), lane under test	—	3.4	dB
OMA _{outer} of each aggressor lane ^e	—	4.2	dBm

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Average receive power, each lane (min) is not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with conformance test signal at TP3 (see 182.8) for the BLER specified in 182.2.

^d These test conditions are for measuring stressed receiver sensitivity. They are not characteristics of the receiver.

^e No aggressors needed for 200GBASE-DR1-2 in a single lane device.

The values for receiver sensitivity (OMA_{outer}), each lane (max) in Table 182–8 vary depending on the value of TECQ for the test source used for measurement. Figure 182–4 illustrates this.

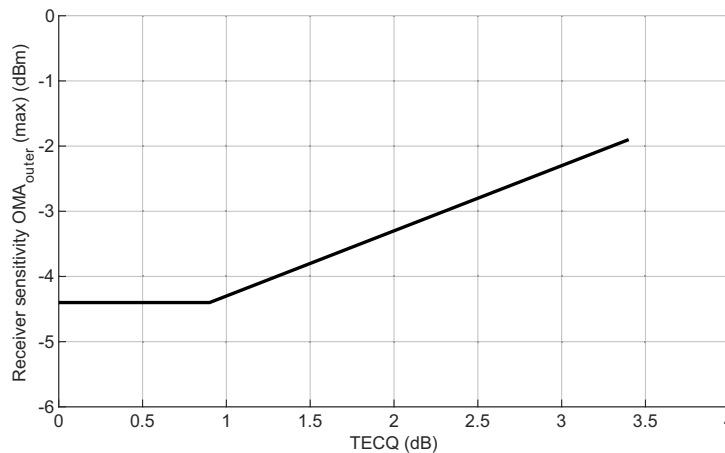


Figure 182-4—Receiver sensitivity (OMA_{outer}), each lane (max)

182.7.3 Illustrative link power budget

An illustrative power budget and penalties for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 channels are shown in Table 182-9.

Table 182-9—Illustrative link power budget

Parameter	Value	Unit
Power budget (for max TDECQ)	7.7	dB
Operating distance	2000	m
Channel insertion loss ^{a, b}	4	dB
Maximum discrete reflectance	−35	dB
Allocation for penalties ^c (for max TDECQ)	3.7	dB
Additional insertion loss allowed	0	dB

^a The channel insertion loss is calculated using the maximum distance specified in Table 182-6 and cabled optical fiber attenuation of 0.5 dB/km at 1304.5 nm plus an allocation for connection and splice loss given in 182.8.2.

^b The channel insertion loss of 4 dB shown in this illustrative link power budget is valid for a maximum MPI penalty of 0.1 dB, which depends on the number and value of discrete reflectances present in the link. This channel insertion loss may be reduced by up to 0.5 dB depending on the discrete reflectances in the link as shown in Table 182-12.

^c Link penalties are used for link budget calculations. They are not requirements and are not meant to be tested. In this illustrative link power budget, the value shown includes an allocation of 0.1 dB for MPI penalty and 0.2 dB for DGD penalty. For cases with a channel insertion loss less than 4 dB, as shown in Table 182-12, the allocation for penalties should be “7.7 – channel insertion loss”.

The values of transmitter OMA_{outer} (max), transmitter OMA_{outer} (min) versus max(TECQ, TDECQ), and receiver sensitivity (OMA_{outer}) (max) versus TECQ are illustrated in Figure 182-5.

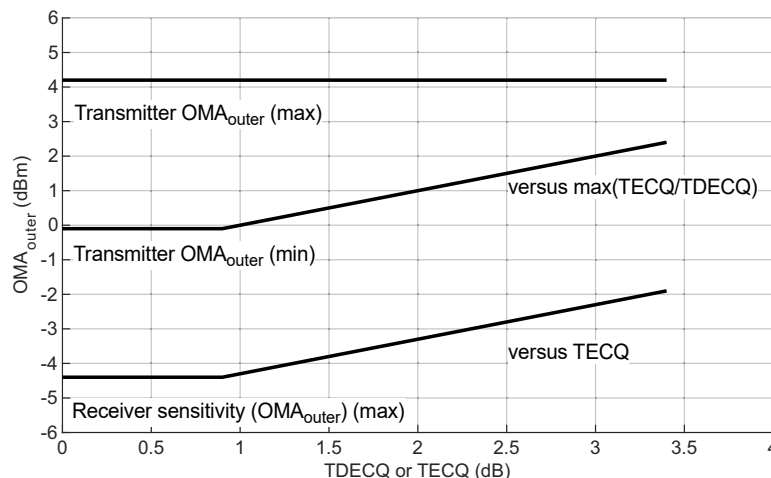


Figure 182-5—Transmitter OMA_{outer} each lane versus $\max(TECQ, TDECQ)$ and receiver sensitivity (OMA_{outer}) each lane versus TECQ

182.8 Optical channel characteristics

The fiber optic cabling model is shown in Figure 182-6.

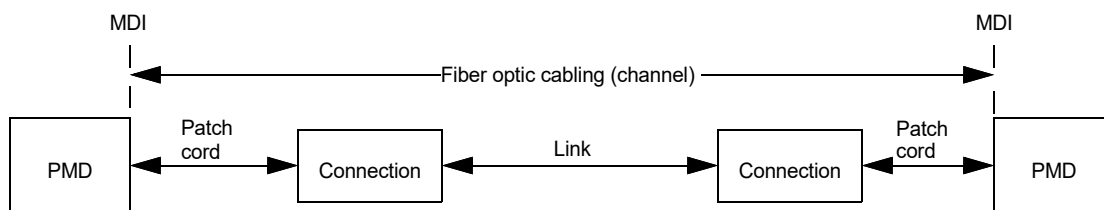


Figure 182-6—Fiber optic cabling model

The optical channel characteristics shall meet the specifications defined in Table 182-10. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together. A channel may contain additional connectors as long as the optical characteristics of the channel, such as attenuation, dispersion, reflections, and polarization mode dispersion meet the specifications. Insertion loss measurements of installed fiber cables are made in accordance with IEC 61280-4 series one-cord reference method. The fiber optic cabling model (channel) defined here applies to each simplex fiber that makes up the duplex fiber link segment. The term channel is used here for consistency with generic cabling standards.

Table 182–10—Optical channel characteristics

Description	200GBASE-DR1-2	400GBASE-DR2-2 800GBASE-DR4-2 1.6TBASE-DR8-2	Unit
Operating distance (max)	2000		m
Channel insertion loss ^{a,b} (max)	See Table 182–12		dB
Channel insertion loss (min)	0		dB
Positive dispersion ^b (max)	2.62		ps/nm
Negative dispersion ^b (min)	–3.41		ps/nm
DGD_max ^c	2.3		ps
Optical return loss (min)	25	37	dB

^a These channel insertion loss values include cable, connectors, and splices.

^b The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024), over the wavelength range 1304.5 nm to 1317.5 nm.

^c Differential Group Delay (DGD) is the time difference at reception between the fractions of a pulse that were transmitted in the two principal states of polarization of an optical signal. DGD_max is the maximum differential group delay that the system is required to tolerate.

182.8.1 Optical fiber cable

The optical fiber cable requirements are satisfied by cables meeting the characteristics in Table 182–11. The use of optical fiber cables containing ITU-T type G.652.D (low water peak, dispersion unshifted), type G.657.A1, or type G.657.A2 (bend insensitive) fibers is recommended.

Table 182–11—Optical fiber and cable characteristics

Description	Value	Unit
Nominal fiber specification wavelength	1310	nm
Cabled optical fiber attenuation (max)	0.5 ^a	dB/km
Zero dispersion wavelength (λ_0)	$1300 \leq \lambda_0 \leq 1324$	nm
Dispersion slope (max) (S_0)	0.092	ps/(nm ² x km)

^a The 0.5 dB/km attenuation is provided for Outside Plant cable as defined in TIA 568-C.3.

182.8.2 Optical fiber connection

An optical fiber connection, as shown in Figure 182–6, consists of a mated pair of optical connectors.

The maximum link distance is based on an allocation of 1 dB total fiber cable attenuation, with the remainder of the maximum channel insertion loss specified in Table 182–12 being allocated to connection

and splice loss. For example, with no discrete reflections greater than -55 dB, a total connection and splice loss allocation of 3 dB supports six connections with an average insertion loss per connection of 0.5 dB. Connections with different loss characteristics may be used provided the requirements of Table 182–10 are met.

Table 182–12—Maximum channel insertion loss versus number of discrete reflectances

Maximum channel insertion loss (dB)		Number of discrete reflectances > −55 dB and ≤ −45 dB								
		0	1	2	3	4	5	6	7	8
Number of discrete reflectances > −45 dB and ≤ −35 dB	0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
	1	4.0	4.0	4.0	4.0	4.0	4.0	4.0	3.9	3.9
	2	4.0	4.0	4.0	3.9	3.9	3.9	3.9	3.9	3.9
	3	3.9	3.9	3.9	3.9	3.8	3.8	3.8	3.8	3.8
	4	3.8	3.8	3.8	3.8	3.8	3.7	3.7	3.7	3.7
	5	3.7	3.7	3.7	3.7	3.7	3.6	3.6	3.6	3.6
	6	3.6	3.6	3.6	3.6	3.5	3.5	3.5	3.5	— ^a

^a The indicated combination of reflectances does not provide a supported maximum channel insertion loss.

182.8.3 Medium Dependent Interface (MDI)

The PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 182–6). The PMD is coupled to the fiber optic cabling through one connector plug into the MDI optical receptacle as shown in Figure 180A–1. Example constructions of the MDI include the following:

- PMD with a connectorized fiber pigtail plugged into an adapter
- PMD receptacle

When the MDI is a connector plug and receptacle connection, it shall meet the interface performance specifications of IEC 61753-1 and IEC 61753-021-02.

NOTE—Transmitter compliance testing is performed at TP2 as defined in 182.5.1, not at the MDI.

Annex 180A specifies the details of the MDIs for 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2.

For 200GBASE-DR1-2, besides the option to connect to an MDI with two fibers, there are two additional specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 400GBASE-DR2-2, and 800GBASE-DR4-2 there are two specified MDI optical receptacles, a single-row 12-fiber interface and a single-row 16 fiber interface.

For 1.6TBASE-DR8-2 there is one specified MDI optical receptacle, a single-row 16 fiber interface.

182.9 Definition of optical parameters and measurement methods

All transmitter optical measurements shall be made through a short patch cable, between 2 m and 5 m in length, unless otherwise specified.

182.9.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. Table 182–14 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 182–14 may be used to perform that test. The test patterns used in this clause are shown in Table 182–13.

Table 182–13—Test patterns

Pattern	Pattern description	Defined in
3	PRBS31Q	177.6.1.4
4	PRBS13Q	177.6.1.3
5	Scrambled idle test pattern encoded by the 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R Inner FEC	119.2.4.9, 172.2.4.11, 175.2.4.11, 177.4
6	SSPRQ	177.6.1.5
7	Valid 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R signal encoded by the 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R Inner FEC	119.2.4, 172.2.4, 175.2.4, 177.4
8	PRBS31 pattern encoded by the 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R Inner FEC	177.6.1.1

Table 182–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Wavelength	3, 4, 5, 6, 7 or 8	182.9.3
Side mode suppression ratio	3, 5, 6, 7 or 8	182.9.3
Average optical power	3, 5, 6, 7 or 8	182.9.4
Outer optical modulation amplitude (OMA _{outer})	4 or 6	182.9.5
Transmitter and dispersion eye closure for PAM4 (TDECQ)	6	182.9.6
Transmitter eye closure for PAM4 (TECQ)	6	182.9.7

Table 182–14—Mapping of parameters to test patterns and related subclauses

Parameter	Pattern	Related subclause
Transmitter functional test	8	182.9.8
Transmitter overshoot and undershoot	6	182.9.9
Transmitter power excursion	6	182.9.10
Extinction ratio	4 or 6	182.9.11
Transmitter transition time	6	182.9.12
RIN _x OMA	4 or 6	182.9.13
Receiver sensitivity	8	182.9.14
Stressed receiver conformance test signal calibration	6	182.9.15
Stressed receiver sensitivity	8	182.9.15

182.9.2 Reference receiver

A reference receiver is defined for transmitter measurements. The reference receiver is composed of an O/E converter, a clock recovery unit (CRU), and a pattern-triggered oscilloscope as shown in Figure 180–7. The combined frequency response of the O/E converter and oscilloscope has a 3 dB bandwidth of approximately 56.71875 GHz with a fourth-order Bessel-Thomson response for frequencies up to at least 1.3 x 113.4375 GHz and a response no higher than –20 dB for frequencies greater than 1.3 x 113.4375 GHz.

The CRU has a corner frequency of 4.27 MHz and a slope of 20 dB/decade. The CRU may be implemented in hardware or software depending on oscilloscope technology.

182.9.3 Wavelength and side mode suppression ratio (SMSR)

The wavelength and SMSR of each optical lane shall be within the ranges given in Table 182–7 if measured per IEC 61280-1-3. The lane under test is modulated using one of the test patterns defined in Table 182–14.

182.9.4 Average optical power

The average optical power of each lane shall be within the limits given in Table 182–7 if measured using the methods given in IEC 61280-1-1. The average optical power is measured using one of the test patterns defined in Table 182–14.

182.9.5 Outer optical modulation amplitude (OMA_{outer})

The OMA_{outer} of each lane shall be within the limits given in Table 182–7. The OMA_{outer} is measured using a test pattern specified for OMA_{outer} in Table 182–14 as the difference between the average optical launch power level P₃, measured over the central 2 UI of a run of 7 threes, and the average optical launch power level P₀, measured over the central 2 UI of a run of 6 zeros, as shown in Figure 182–7. OMA_{outer} is measured using waveforms captured at the output of the reference receiver defined in 182.9.2.

When measured with pattern 6, OMA_{outer} is measured as the difference between the average optical launch power level P₃, measured over the central 2 UI of the first 7 UI of all runs of 7 or more threes, and the average optical launch power level P₀, measured over the central 2 UI of the first 6 UI of all runs of 6 or more zeros.

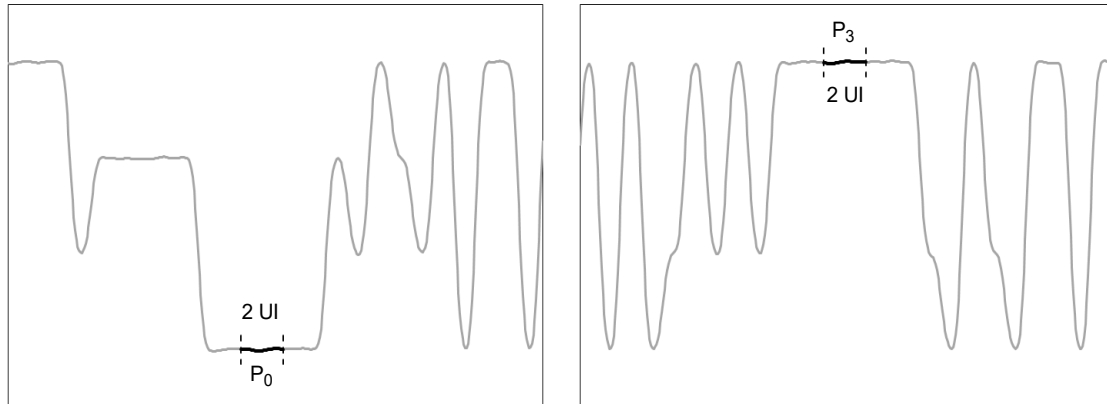


Figure 182-7—Example power levels P_0 and P_3 from PRBS13Q test pattern

182.9.6 Transmitter and dispersion eye closure for PAM4 (TDECQ)

The TDECQ of each lane shall be within the limit given in Table 182-7. The TDECQ is measured using the methods specified in 180.9.6 with the following exceptions:

- The signaling rate of the test pattern generator is as given in Table 182-7 and uses the test pattern specified for TDECQ in Table 182-14.
- The reference receiver is as specified in 182.9.2
- The target PAM4 symbol error ratio is 9.6×10^{-3} and the related Q_t value is 2.489.
- The normalized noise power density spectrum, $N(f)$ is equivalent to white noise filtered by a fourth-order Bessel-Thomson response filter with a 3 dB bandwidth of 56.71875 GHz
- The clock recovery unit (CRU) has a corner frequency of 4.27 MHz.
- The optical return loss is as given in Table 182-7.
- The aggressor lanes shall have power levels as specified in Table 182-8 for the stressed receiver sensitivity test.
- For those cases where there is an xAUI-n chip-to-chip (C2C) or chip-to-module (C2M) interface (see Table 182-1 through Table 182-4), the clock source for the SSPRQ test pattern is derived from the clock recovered from the xAUI-n input signal. The pattern of the xAUI-n input signal may be PRBS31Q, scrambled idle, or a valid xBASE-R signal.
- The transmitter is tested using a worst case optical channel specified in 182.9.6.1.

182.9.6.1 Channel requirements

The transmitter is tested using an optical channel that meets the requirements listed in Table 182-15.

Table 182-15—Transmitter compliance channel specifications

Dispersion ^a (ps/nm)		Insertion loss ^b	Optical return loss ^c (dB)	Max mean DGD ^d (ps)
Minimum	Maximum			
$0.1850 \times (\lambda - 1311) - 2.22$	$0.1770 \times (\lambda - 1311) + 1.47$	Minimum	See Table 182-7	0.8

- ^a The dispersion is measured for the wavelength of the transmitter lane under test (λ in nm). The coefficient assumes 2 km distance. The dispersion specifications are based on the statistical link design methodology documented in Appendix I of Recommendation ITU-T G.652 (08/2024).
- ^b The value “Minimum” implies that the test channel insertion loss does not excessively degrade the signal to noise ratio of the measurement.
- ^c The back-reflection is applied at TP2.
- ^d The limit for maximum mean DGD is to ensure that the contribution to TDECQ from DGD is substantially less than the primary contribution from the channel due to chromatic dispersion.

A transmitter is to be compliant with a total dispersion at least as negative as the “minimum dispersion” and at least as positive as the “maximum dispersion” columns specified in Table 182–15 for the wavelength of the transmitter lane under test. This may be achieved with channels consisting of fibers with lengths chosen to meet the dispersion requirements.

To verify that the fiber has the correct amount of dispersion, the measurement method defined in IEC 60793-1-42 may be used. The measurement is made in the linear power regime of the fiber.

The channel provides back reflection with return loss specified in Table 182–15. The state of polarization of the back reflection is adjusted to create the greatest RIN.

The mean DGD of the channel is to be less than the value specified in Table 182–15.

182.9.7 Transmitter eye closure for PAM4 (TECQ)

The TECQ of each lane is a measure of the optical transmitter eye closure at TP2. The TECQ shall be within the limit given in Table 182–7 if measured using a test pattern specified for TECQ in Table 182–14. TECQ is measured using the methods specified for TDECQ in 182.9.6, except that the test fiber is not used.

182.9.8 Transmitter functional test

The transmitter functional symbol error histogram is measured using the method defined in 180.9.8 with the following exceptions:

- The transmitter under test is connected to the functional receiver (FnRx) by a test fiber which meets the requirements in 182.9.6.1. Full compliance with 182.9.6.1 requires the test fiber to provide back reflection with polarization state resulting in the greatest RIN. Other test fibers, including patch cords, may be used for additional tests.

where in 180.9.8.1

- The FnRx is a variable optical attenuator (VOA) followed by an optical receiver (ORx) that complies with characteristics as given in Table 182–8. The VOA and test fiber approximate the compliance channel as given in Table 182–15 with the exception that the channel loss is not minimum.

where in Equation (180–16)

- $Tx_DUT_power_budget$ is the transmitter under test power budget for the specific fiber under test as given in Table 182–9, except uses measured instead of max TDECQ value and is given by Equation (180–17).

where in Equation (180–17)

- $Channel_insertion_loss$ is the “Channel insertion loss” as given in Table 182–9.
- $MPI_and_DGD_penalties_allocation$ is the allocation for MPI and DGD penalties as given in Table 182–9.

where in 180.9.8.2

- The transmitter functional symbol error histogram mask per each lane is given in Table 180–17 and is measured using the test pattern as given in Table 182–14.

182.9.9 Transmitter overshoot and undershoot

The transmitter overshoot and undershoot of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for overshoot and undershoot in Table 182–14.

Transmitter overshoot and undershoot are measured using the waveform captured for the TDECQ test (see 182.9.6) and the waveform captured for the TECQ test (see 182.9.7), at the output of the reference receiver defined in 182.9.2, before the reference equalizer.

Transmitter overshoot and undershoot are calculated using the methods in 140.7.7, with the exceptions that P_0 , P_3 and OMA_{outer} are measured using the methods in 182.9.5 and the hit ratio used to determine P_{max} and P_{min} is 1×10^{-4} .

182.9.10 Transmitter power excursion

The transmitter power excursion of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for transmitter power excursion in Table 182–14.

Transmitter power excursion is measured using the waveforms captured at the output of the reference receiver defined in 182.9.2.

Transmitter power excursion is defined in 140.7.8, with the exception that the hit ratio used to determine P_{max} and P_{min} is 1×10^{-4} .

182.9.11 Extinction ratio

The extinction ratio of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for extinction ratio in Table 182–14. The extinction ratio of a PAM4 optical signal is defined as the ratio of the average optical launch power levels P_3 and P_0 , as defined for OMA_{outer} (see 182.9.5).

182.9.12 Transmitter transition time

The transmitter transition time of each lane shall be within the limit given in Table 182–7 if measured using a test pattern specified for transmitter transition time in Table 182–14.

Transmitter transition time is defined as the slower of the time interval of the transition from 20% of OMA_{outer} to 80% of OMA_{outer} , or from 80% of OMA_{outer} to 20% of OMA_{outer} , for the rising and falling edges respectively. The transmitter transition time is measured using the waveforms captured at the output of the reference receiver defined in 182.9.2.

The 0% level and the 100% level are P_0 and P_3 as defined by the OMA_{outer} measurement procedure (see 182.9.5). When the SSPRQ pattern is used, the rising edge used for the measurement is that within the 00000333333 symbol sequence, and the falling edge is that within the 33333000000 symbol sequence.

182.9.13 Relative intensity noise (RIN_xOMA)

RIN_xOMA of each lane, with “x” referring to the value for optical return loss tolerance in Table 182–7, shall be within the limit given in Table 182–7 when measured using the test pattern and sampling range specified for OMA_{outer} measurement in 182.9.5, but with applied x dB optical return loss and the reference receiver specified in 182.9.2. RIN_xOMA is measured using the methods specified in 180.9.13.

182.9.14 Receiver sensitivity

The receiver sensitivity (OMA_{outer}) of each lane shall be within the limits given in Table 182–8 if measured using a test pattern for receiver sensitivity in Table 182–14. The conformance test signal at TP3 meets the requirements for a transmitter followed by an attenuator.

The TECQ of the conformance test signal is measured according to 182.9.6, except that the test fiber is not used. The measured value of TECQ is then used to calculate the limit for receiver sensitivity (OMA_{outer}) as specified in Table 182–8.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

The measured receiver sensitivity is the lowest value of OMA_{outer} where the receiver meets the BLER requirements in 182.2, measured at the PMA using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 182–16. The error mask $H_{max}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

The Low-stressed Receiver Sensitivity (LRS) conformance test signal at TP3 meets the transmit characteristics given in Table 182–7 except for characteristics in Table 180–18.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_m(k)) - \log_{10}(H_m(4))) / (k - 4)$ should be less than $(\log_{10}(H_m(4)) - \log_{10}(H_m(2))) / 2$ for each k greater than 4.

Table 182–16—Error ratio parameters

Parameter name	Value
p	
200GBASE-DR1-2	1
400GBASE-DR2-2	2
800GBASE-DR4-2	4
1.6TBASE-DR8-2	8
BLER limit	1.45×10^{-11}
BER_{total}	2.921×10^{-4}
BER_{added}	6.4×10^{-5}
BER_{max}	2.281×10^{-4}

182.9.15 Stressed receiver sensitivity

Stressed receiver sensitivity of each lane shall be within the limit given in Table 182–8 if measured using the method defined in 121.8.10 with the following exceptions:

- The SECQ of the stressed receiver conformance test signal is measured according to 182.9.6, except that the test fiber is not used. The transition time of the stressed receiver conformance test signal is no greater than the value specified in Table 182–7.

- With the Gaussian noise generator on and the sinusoidal jitter and sinusoidal interferer turned off, the RIN_x OMA of the SRS test source should be no greater than the value specified in Table 182–7.
- The frequency of the sinusoidal interferer is greater than 200 MHz.
- The signaling rate of the test pattern generator and the extinction ratio of the E/O converter are as given in Table 182–7 using test patterns specified in Table 182–14.
- The stressed receiver conformance input signal meets the sinusoidal jitter specified in 182.9.15.1.
- The required values of the “Stressed receiver sensitivity (OMA_{outer}), each lane (max)”, “Stressed eye closure for PAM4 (SECQ), lane under test” and “OMA_{outer} of each aggressor lane” are as given in Table 182–8.

The measured stressed receiver sensitivity is the lowest value of OMA_{outer} where the receiver meets the requirements in 182.2 using the test method in either 174A.9.5 or 174A.9.6 with parameters provided in Table 182–16. The error mask $H_{max}(k)$ to be used in the method of 174A.9.5 is provided in Table 180–20. When testing using 174A.9.5 or 174A.9.6, the compliance shall not be further verified using 174A.9.7.

Precoding (see 176.7.1.2) is enabled if the receiver requests precoding using the ILT function.

NOTE—In order to predict whether a receiver meets the BLER requirement in a short test time, extrapolation of the measured histogram (see 174A.9.3) to $H_m(16)$ can be performed. One way of doing this is using a linear fit of $\log_{10}(H_m(k))$ for $k = (n - 2)$ to n , where these values of k are the three highest bins having a count of 10 or more. If the slope of the histogram over the measured bins is increasing, it might be an indication of not meeting the BLER requirement. To verify that the slope is not increasing, the values of $(\log_{10}(H_m(k)) - \log_{10}(H_m(4))) / (k - 4)$ should be less than $(\log_{10}(H_m(4)) - \log_{10}(H_m(2))) / 2$ for each k greater than 4.

182.9.15.1 Sinusoidal jitter for receiver conformance test

The sinusoidal jitter is used to test receiver jitter tolerance. The amplitude of the applied sinusoidal jitter is dependent on frequency as specified in Table 182–17 and is illustrated in Figure 182–8.

Table 182–17—Applied sinusoidal jitter

Frequency range	Sinusoidal jitter, peak-to-peak (UI)
$f < 42.7$ kHz	Not specified
42.7 kHz $< f \leq 4.27$ MHz	2.13×10^5 Hz/ f
4.27 MHz $< f < 10$ LB ^a	0.05

^a LB = loop bandwidth; upper frequency bound for added sine jitter should be at least 10 times the loop bandwidth of the receiver being tested.

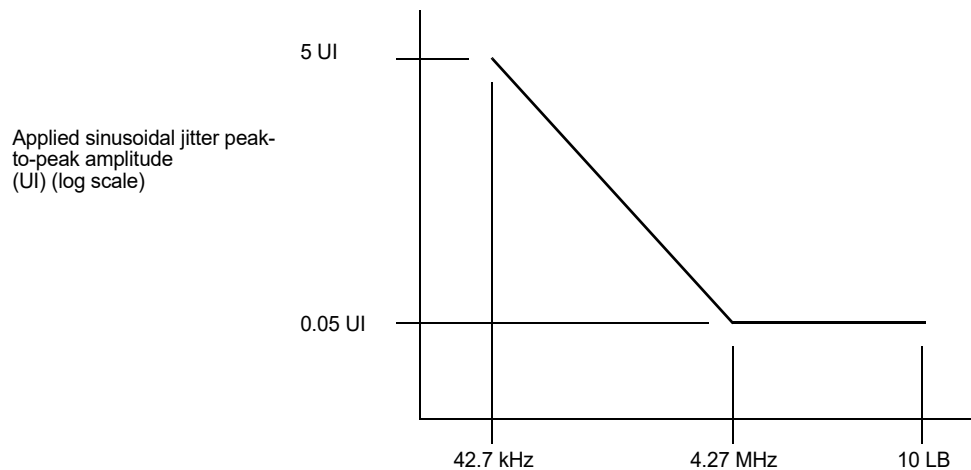


Figure 182-8—Illustration of the mask of the sinusoidal component of jitter tolerance

182.10 Safety, installation, environment, and labeling

182.10.1 General safety

All equipment subject to this clause shall conform to IEC 63268-1.

182.10.2 Laser safety

200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2 optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product's laser, safety features, labeling, use, maintenance, and service. This documentation explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.¹³

182.10.3 Installation

It is recommended that proper installation practices, as defined by applicable local codes and regulation, be followed in every instance in which such practices are applicable.

¹³A host system that fails to meet the manufacturer's requirements and/or usage restrictions may emit laser radiation in excess of the safety limits of one or more safety standards. In such a case, the host manufacturer is required to obtain its own laser safety certification.

182.10.4 Environment

Normative specifications in this clause shall be met by a system integrating a 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 PMD over the life of the product while the product operates within the manufacturer's range of environmental, power, and other specifications.

It is recommended that manufacturers indicate in the literature associated with the PHY the operating environmental conditions to facilitate selection, installation, and maintenance.

It is recommended that manufacturers indicate, in the literature associated with the components of the optical link, the distance and operating environmental conditions over which the specifications of this clause will be met.

182.10.5 Electromagnetic emission

A system integrating a 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, or 1.6TBASE-DR8-2 PMD shall comply with applicable local and national codes for the limitation of electromagnetic interference.

182.10.6 Temperature, humidity, and handling

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

182.10.7 PMD labeling requirements

It is recommended that each PHY (and supporting documentation) be labeled in a manner visible to the user, with at least the applicable safety warnings and the applicable port type designation (e.g., 800GBASE-DR4-2).

Labeling requirements for Hazard Level 1 lasers are given in the laser safety standards referenced in 182.10.2.

182.11 Management variables

PMD control and status variables intended to be accessible via a management system are listed in Table 182–18 and Table 182–19. Additional variables associated with the APSU functions are listed in Table 178B–3 and Table 178B–4.

If the MDIO interface (see Clause 45) is implemented, PMD control and status variables shall be mapped to MDIO registers as shown in Table 182–18 and Table 182–19.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 182–18—PMD control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_reset	182.5.6	1.0.15	45.2.1.1
Global_PMD_transmit_disable	182.5.7	1.9.0	45.2.1.8
PMD_transmit_disable_7 to PMD_transmit_disable_4 ^a	182.5.8	1.9.8 to 1.9.5	45.2.1.8
PMD_transmit_disable_3, PMD_transmit_disable_2 ^b	182.5.8	1.9.4, 1.9.3	45.2.1.8
PMD_transmit_disable_1 ^c	182.5.8	1.9.2	45.2.1.8
PMD_transmit_disable_0	182.5.8	1.9.1	45.2.1.8

^aAvailable only in 1.6TBASE-DR8-2

^bAvailable only in 800GBASE-DR4-2 and 1.6TBASE-DR8-2

^cAvailable only in 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

Table 182–19—PMD status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PMD_fault	182.5.9	1.1.7	45.2.1.2
PMD_transmit_fault	182.5.9	1.8.11	45.2.1.7
PMD_receive_fault	182.5.11	1.8.10	45.2.1.7
Global_PMD_signal_detect	182.5.4	1.10.0	45.2.1.9
PMD_signal_detect_7 to PMD_signal_detect_4 ^a	182.5.5	1.10.8 to 1.10.5	45.2.1.9
PMD_signal_detect_3, PMD_signal_detect_2 ^b	182.5.5	1.10.4, 1.10.3	45.2.1.9
PMD_signal_detect_1 ^c	182.5.5	1.10.2	45.2.1.9
PMD_signal_detect_0	182.5.5	1.10.1	45.2.1.9

^aAvailable only in 1.6TBASE-DR8-2

^bAvailable only in 800GBASE-DR4-2 and 1.6TBASE-DR8-2

^cAvailable only in 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

182.12 Protocol implementation conformance statement (PICS) proforma for Clause 182, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2¹⁴

182.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 182, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

182.12.2 Identification

182.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

182.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-20xx, Clause 182, Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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¹⁴Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

182.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*DR12	200GBASE-DR1-2	182.7	Device supports requirements for 200GBASE-DR1-2 PHY	O.1	Yes [] No []
*DR22	400GBASE-DR2-2	182.7	Device supports requirements for 400GBASE-DR2-2 PHY	O.1	Yes [] No []
*DR42	800GBASE-DR4-2	182.7	Device supports requirements for 800GBASE-DR4-2 PHY	O.1	Yes [] No []
*DR82	1.6TBASE-DR8-2	182.7	Device supports requirements for 1.6TBASE-DR8-2 PHY	O.1	Yes [] No []
*INS	Installation / cable	182.8	Items marked with INS include installation practices and cable specifications not applicable to a PHY manufacturer	O	Yes [] No []
DC	Delay constraints	182.4.1	Device conforms to delay constraints	M	Yes []
SC	Skew constraints	182.4.2	Device conforms to Skew and Skew Variation constraints	M	Yes []
*MD	MDIO capability	182.11	Registers and interface supported	O	Yes [] No []

182.12.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2

182.12.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
F1	Compatible with 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R PCS and PMA	182.1		M	Yes []
F2	Block error ratio	182.2	Meets the BLER specified in 182.2	M	Yes []
F3	Transmit function	182.5.2	Conveys symbols from PMD service interface to MDI	M	Yes []
F4	Mapping between optical signal and logical signal for transmitter	182.5.2	Optical power levels from lowest to highest correspond to tx_symbols zero, one, two, and three, respectively	M	Yes []
F5	Receive function	182.5.3	Conveys symbols from MDI to PMD service interface	M	Yes []
F6	Conversion of four optical signals to symbol streams	182.5.3	For delivery to the PMD service interface	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
F7	Mapping between optical signal and logical signal for receiver	182.5.3	Optical power levels from lowest to highest correspond to rx_symbols zero, one, two, and three, respectively	M	Yes []
F8	Global Signal Detect behavior	182.5.4	Global_PMD_signal_detect is an indicator of the presence of an optical signal on all lanes	M	Yes []
F9	Lane-by-lane Signal Detect function	182.5.5	Sets PMD_signal_detect_i values on a lane-by-lane basis per requirements of Table 182–5	M	Yes []
F10	PMD reset function	182.5.6	Resets the PMD sublayer	O	Yes [] No []
F11	Global transmit disable function	182.5.7	Disables all of the optical transmitters with the Global_PMD_transmit_disable variable	O	Yes [] No []
F12	PMD lane-by-lane transmit disable using management variable	182.5.8	Disables the optical transmitter on the lane associated with the PMD_transmit_disable_i variable	O	Yes [] No []
F13	PMD lane-by-lane transmit disable using state variable	182.5.8	Disables the optical transmitter when tx_disable variable is true	M	Yes []
F14	PMD_fault function	182.5.9	Sets PMD_fault to one if any local fault is detected	O	Yes [] No []
F15	PMD_transmit_fault function	182.5.10	Sets PMD_transmit_fault to one if a local fault is detected on any transmit lane	O	Yes [] No []
F16	PMD_receive_fault function	182.5.11	Sets PMD_receive_fault to one if a local fault is detected on any receive lane	O	Yes [] No []

182.12.4.2 PMD to MDI optical specifications for 200GBASE-DR1-2

Item	Feature	Subclause	Value/Comment	Status	Support
DDR1	Transmitter meets specifications in Table 182–7	182.7.1	Per definitions in 182.9	DR12:M	Yes [] N/A []
DDR2	Receiver meets specifications in Table 182–8	182.7.2	Per definitions in 182.9	DR12:M	Yes [] N/A []

182.12.4.3 PMD to MDI optical specifications for 400GBASE-DR2-2

Item	Feature	Subclause	Value/Comment	Status	Support
EDR1	Transmitter meets specifications in Table 182–7	182.7.1	Per definitions in 182.9	DR22:M	Yes [] N/A []
EDR2	Receiver meets specifications in Table 182–8	182.7.2	Per definitions in 182.9	DR22:M	Yes [] N/A []

182.12.4.4 PMD to MDI optical specifications for 800GBASE-DR4-2

Item	Feature	Subclause	Value/Comment	Status	Support
FDR1	Transmitter meets specifications in Table 182–7	182.7.1	Per definitions in 182.9	DR42:M	Yes [] N/A []
FDR2	Receiver meets specifications in Table 182–8	182.7.2	Per definitions in 182.9	DR42:M	Yes [] N/A []

182.12.4.5 PMD to MDI optical specifications for 1.6TBASE-DR8-2

Item	Feature	Subclause	Value/Comment	Status	Support
GDR1	Transmitter meets specifications in Table 182–7	182.7.1	Per definitions in 182.9	DR82:M	Yes [] N/A []
GDR2	Receiver meets specifications in Table 182–8	182.7.2	Per definitions in 182.9	DR82:M	Yes [] N/A []

182.12.4.6 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Measurement cable	182.9	2 m to 5 m in length	M	Yes []
OM2	Center wavelength and SMSR	182.9.3	Per IEC 61280-1-3 under modulated conditions	M	Yes []
OM3	Average optical power	182.9.4	Per IEC 61280-1-1	M	Yes []
OM4	OMA measurements	182.9.5	Each lane	M	Yes []
OM5	Transmitter and dispersion eye closure for PAM4 (TDECQ)	182.9.6	Each lane	M	Yes []
OM6	Transmitter eye closure for PAM4 (TECQ)	182.9.7	Each lane	M	Yes []
OM8	Transmitter functional symbol error histogram	182.9.8	Each lane	M	Yes []
OM8	Transmitter overshoot and undershoot	182.9.9	Each lane	M	Yes []
OM9	Transmitter power excursion	182.9.10	Each lane	M	Yes []
OM10	Extinction ratio	182.9.11	Each lane	M	Yes []
OM11	Transmitter transition time	182.9.12	Each lane	M	Yes []
OM12	Relative intensity noise	182.9.13	Each lane	M	Yes []
OM13	Receiver sensitivity	182.9.14	Each lane	M	Yes []
OM14	Stressed receiver sensitivity	182.9.15	Each lane	M	Yes []

182.12.4.7 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	182.10.1	Conforms to J.2	M	Yes []
ES2	Laser safety—IEC Hazard Level 1	182.10.2	Conforms to Hazard Level 1 laser requirements defined in IEC 60825-1 and IEC 60825-2	M	Yes []
ES3	Electromagnetic interference	182.10.5	Complies with applicable local and national codes for the limitation of electromagnetic interference	M	Yes []

182.12.4.8 Characteristics of the fiber optic cabling and MDI

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	Fiber optic cabling	182.8	Meets requirements specified in Table 182–10 and Table 182–11	INS:M	Yes [] N/A []
OC2	Maximum discrete reflectance	182.8.2	Meets requirements specified in Table 182–12	INS:M	Yes [] N/A []
OC3	MDI Requirements	182.8.3	Meets IEC 61753-1 and IEC 61753-021-2	INS:M	Yes [] N/A []

182.12.4.9 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	182.11		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	182.11		MD:M	Yes [] N/A []